

MVC Architecture

M - Model

V - View

C - Controller

MVC is design pattern that separates application into layers so that code is easy to understand, maintain, test and scale.

Model - Data + Business Logic

- handles data
- talks to the database
- contains business rules

Examples - Database schemas, Data validation logic

Model does not care about the HTTP method it receives

View - UI / Presentation

- display data to the user
- formats output

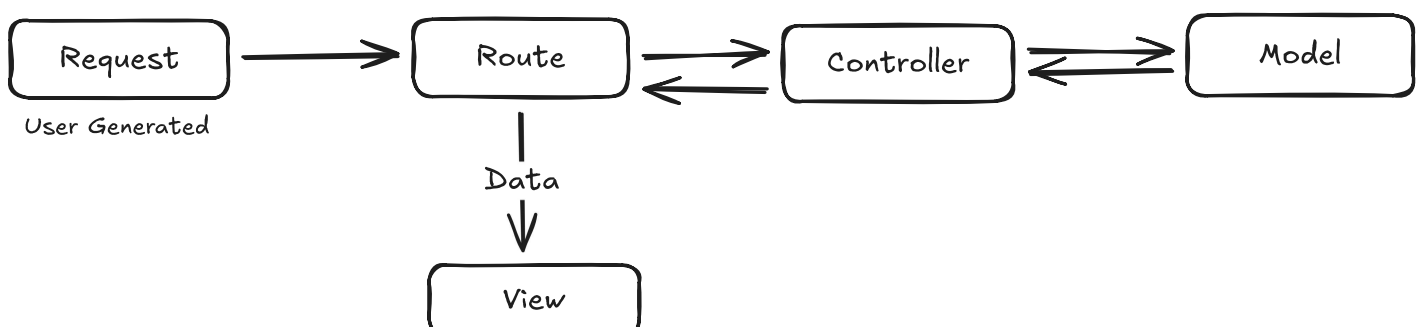
in Node.js views can be

1. template engines - ejs, handlebars or pug
2. JSON responses
3. Frontend Frameworks - React, Angular or Vue (Headless Architecture)

Controller - Request Handler

- receives HTTP request
- calls the Model
- sends the data to the view

Controllers are mediator between request and the data



Setup a Node Application

npm init - insert all the details asked in prompt

npm i express - to create express server

npm i nodemon - to auto-restart the server if any change is detected

Mongoose

Model - is the layer where the actual business logic and schema is defined

ODM - Object Data Model - for NoSQL

ORM - Object Relation Model - for SQL

Mongoose is an ODM library for MongoDB and Node.js

Mongoose helps in working of Node.js application with MongoDB in structured, JS- friendly way

Why Mongoose is Used?

1. Schema Definition

- we can define a schema for a collection to make data stay consistent.

2. Models - Database interaction

- Schemas are compiled into Models, which we use to interact with the database

3. Build-in Validation

- Mongoose validates data before saving into MongoDB
- if validation fails, database is not updated

4. Middlewares

- Running code before or after database operations

npm i mongoose